OPTIMIND

Ubuntu + Docker Setup

A practical guide to running the local-first AI workspace on an Ubuntu server.

Includes PostgreSQL, Ollama, SearXNG, FastAPI, and the React frontend.

October 2026 • Internal deployment guide

1. What this setup runs

The Docker Compose profile starts the core OptiMind stack as separate containers:

- React/Vite frontend on port 5173
- FastAPI backend on port 8000
- PostgreSQL with persistent database storage
- Ollama with a persistent model volume
- SearXNG for free local web search

Database migrations run automatically when the backend container starts. The first run downloads the configured Ollama model and may take several minutes.

Image generation is disabled by default because its CPU/GPU runtime is hardware-specific. It can be added separately after the core stack is working.

2. Ubuntu prerequisites

Install Ubuntu updates, Git, Docker Engine, and the Docker Compose plugin. Confirm the installation:

```
docker --version
docker compose version
git --version
```

If Docker requires sudo, either prefix Docker commands with sudo or add your user to the Docker group and sign in again:

```
sudo usermod -aG docker "$USER"
```

3. Download and configure

```
git clone <your-repository-url> optimind
cd optimind
cp .env.docker.example .env
chmod 600 .env
```

Edit .env and set a strong database password and search secret. Choose the model that fits the server:

Server memory	Suggested model	Trade-off
8 GB	qwen2.5:1.5b	Fastest and lightest
16 GB+	qwen2.5:3b	Better quality

```
OLLAMA_MODEL=qwen2.5:1.5b
```

4. Start the project

```
docker compose up -d --build
docker compose ps
```

Open the app on the Ubuntu machine:

```
http://localhost:5173
```

Follow startup logs or check all dependencies:

```
docker compose logs -f backend
curl http://localhost:8000/api/v1/status
```

5. Automatic startup after reboot

The services use restart: unless-stopped. Enable Docker at boot and start the stack once:

```
sudo systemctl enable --now docker
docker compose up -d
```

After Ubuntu reboots, Docker automatically brings the containers back. Verify with:

```
docker compose ps
```

6. Access from another office computer

The frontend is published on all host interfaces. Find the Ubuntu server's static/private IP:

```
hostname -I
```

If the address is 192.168.1.50, open this from another computer on the same office network:

```
http://192.168.1.50:5173
```

If UFW is enabled, allow only the private office subnet:

```
sudo ufw allow from 192.168.1.0/24 to any port 5173 proto tcp
sudo ufw status
```

Replace 192.168.1.0/24 with the actual office network range.

> **Do not expose PostgreSQL, Ollama, SearXNG, or port 8000 to the public internet. Only expose the frontend, or place it behind a VPN/reverse proxy.**

7. Use an office hostname

Yes. Create an internal DNS record in the office router, Pi-hole, or Windows/AD DNS:

```
optimind.office.example  ->  192.168.1.50
```

Then add the hostname to .env:

```
VITE_ALLOWED_HOSTS=optimind.office.example
```

Rebuild the frontend:

```
docker compose up -d --build frontend
```

Users can then open:

```
http://optimind.office.example:5173
```

For a quick office-only test without DNS, add this line to each client computer's hosts file:

```
192.168.1.50  optimind.office
```

Then browse to http://optimind.office:5173.

8. Stop, update, and troubleshoot

```
docker compose down                  # stop without deleting data
docker compose up -d --build         # rebuild after code changes
docker compose logs -f                # follow all logs
docker compose restart backend       # restart only the API
```

To remove the database and Ollama model volumes as well (destructive):

```
docker compose down -v
```

If the page does not open, check the container state, listening port, firewall, and server address:

```
docker compose ps
ss -ltnp | grep 5173
curl http://192.168.1.50:5173
```

For a polished permanent deployment, put Nginx or Caddy in front of the frontend and use HTTPS. Do not expose the Vite development server directly to the public internet.